Cloud-containerization-and-DevOps

Experiment 9: Working with Ansible

Introduction

What is Ansible?

- Ansible is an open-source **automation tool** for **configuration management**, **application deployment**, and **orchestration**.
- It follows an **agentless** architecture, using SSH for Linux and WinRM for Windows.
- Uses YAML-based **playbooks** to define automation tasks.

> Ansible is software that enables cross-platform automation and orchestration at scale and has become the standard choice among enterprise automation solutions.

Key Concepts

| Component | Description | | :————————-|:————————| | **Control Node**| Machine with Ansible installed| | **Managed Nodes**| Target servers (no Ansible agent needed)| | **Inventory** | Defines the list of managed nodes (EC2 instances, servers, etc.). `inventory.ini` | | **Playbooks** | YAML files containing a sequence of automation steps. | | **Tasks** | Individual actions in playbooks (e.g., installing a package). | | **Modules** | Built-in functionality to perform tasks (e.g., `yum` , `apt` , `service`). | | **Roles** | Pre-defined reusable automation scripts. |

Benefits of using Ansible

- A free and open-source community project with a huge audience.
- Battle-tested over many years as the preferred tool of IT wizards.
- Easy to start and use from day one, without the need for any special coding skills.
- Simple deployment workflow without any extra agents.
- Includes some sophisticated features around modularity and reusability that come in handy as users become more proficient.
- Extensive and comprehensive official documentation that is complemented by a plethora of online material produced by its community.

> To sum up, Ansible is simple yet powerful, agentless, community-powered, predictable, and secure.

Hands-On

Step-1:- Install Ansible in WSL

```
sudo apt install -y
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ sudo apt install ansible -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ansible-core ieee-data python3-argcomplete python3-dnspython
  python3-jmespath python3-kerberos python3-libcloud python3-lockfile
  python3-netaddr python3-ntlm-auth python3-packaging python3-passlib
  python3-requests-ntlm python3-resolvelib python3-selinux
  python3-simplejson python3-winrm python3-xmltodict
Suggested packages:
  cowsay sshpass python3-trio python3-aioquic python3-h2 python3-httpx
  python3-httpcore python-lockfile-doc ipython3 python-netaddr-docs
The following NEW packages will be installed:
  ansible ansible-core ieee-data python3-argcomplete python3-dnspython
  python3-jmespath python3-kerberos python3-libcloud python3-lockfile
  python3-netaddr python3-ntlm-auth python3-packaging python3-passlib
  python3-requests-ntlm python3-resolvelib python3-selinux
  python3-simplejson python3-winrm python3-xmltodict
0 upgraded, 19 newly installed, 0 to remove and 91 not upgraded.
Need to get 22.0 MB of archives.
After this operation, 330 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-packaging al
l 24.0-1 [41.1 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 python3-resolvel
ib all 1.0.1-1 [25.7 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-dnspython al
l 2.6.1-1ubuntu1 [163 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 ieee-data all 202208
27.1 [2113 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-netaddr all
0.8.0-2ubuntu1 [319 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 ansible-core all
```

Step-2:- Verify Installation

```
ansible --version
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/palakagarwal_16/.ansible/plugins/m
odules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/palakagarwal_16/.ansible/collections:/
usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Jan 22 2026, 20:57:42) [GCC 13.3.0] (/usr/b
in/python3)
  jinja version = 3.1.2
  libyaml = True
```

Step-3:- Ping Localhost

```
ansible localhost -m ping
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ ansible localhost -m ping
[WARNING]: No inventory was parsed, only implicit localhost is available
localhost | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
```

Step-4:- Generate `ssh` Keys

```
ssh -keygen -t rsa -b 4096
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/palakagarwal_16/.ssh/id_rsa): Sem
6
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in Sem6
Your public key has been saved in Sem6.pub
The key fingerprint is:
SHA256:oFx17tPdckMM+BiN7SyFn8TM2FbA5Evj7Sz8uqcVZ2c palakagarwal_16@LAPTOP-2S
DBVCPO
The key's randomart image is:
+---[RSA 4096]----+
|        . . #+o. |
|       . o *.%o  |
|      o   . @+.o |
|   . o . . +oB=. |
|    o   S o ooo+E|
|           .. o==|
|             o + |
|              +. |
|             +=. |
+----[SHA256]-----+
```

Step-5:- Copy keys in `id_rsa`

```
#Copy in current Directory
cp ~/.ssh/id_rsa.pub .
cp ~/.ssh/id_rsa .
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/ansible-lab$ cp ~/.ssh/id_rs
a.pub .
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/ansible-lab$ cp ~/.ssh/id_rs
a .
```

Step-6:- Create Dockerfile

```
FROM ubuntu


RUN apt update -y
RUN apt install -y python3 python3-pip openssh-server
RUN mkdir -p /var/run/sshd

# Configure SSH
RUN mkdir -p /run/sshd && \
    echo 'root:password' | chpasswd && \
    sed -i 's/#PermitRootLogin prohibit-password/PermitRootLogin yes/' /etc/ssh/sshd_confi
    sed -i 's/#PasswordAuthentication yes/PasswordAuthentication no/' /etc/ssh/sshd_config
    sed -i 's/#PubkeyAuthentication yes/PubkeyAuthentication yes/' /etc/ssh/sshd_config


# Create .ssh directory and set proper permissions
RUN mkdir -p /root/.ssh && \
    chmod 700 /root/.ssh

# Copy SSH keys (note: this is not secure for production!)
COPY id_rsa /root/.ssh/id_rsa
COPY id_rsa.pub /root/.ssh/authorized_keys

# Set proper permissions for keys
RUN chmod 600 /root/.ssh/id_rsa && \
    chmod 644 /root/.ssh/authorized_keys

# Fix for SSH login
RUN sed -i 's@session\s*required\s*pam_loginuid.so@session optional pam_loginuid.so@g' /et

# Expose SSH port
EXPOSE 22

# Start SSH service when container starts
CMD ["/usr/sbin/sshd", "-D"]
```

```
  GNU nano 7.2                                                    Dockerfile
FROM ubuntu

RUN apt update -y
RUN apt install -y python3 python3-pip openssh-server
RUN mkdir -p /var/run/sshd

# Configure SSH
RUN mkdir -p /run/sshd && \
    echo 'root:password' | chpasswd && \
    sed -i 's/#PermitRootLogin prohibit-password/PermitRootLogin yes/' /etc/ssh/sshd_config && \
    sed -i 's/#PasswordAuthentication yes/PasswordAuthentication no/' /etc/ssh/sshd_config && \
    sed -i 's/#PubkeyAuthentication yes/PubkeyAuthentication yes/' /etc/ssh/sshd_config

# Create .ssh directory
RUN mkdir -p /root/.ssh && \
    chmod 700 /root/.ssh

# Copy SSH keys
COPY id_rsa /root/.ssh/id_rsa
COPY id_rsa.pub /root/.ssh/authorized_keys

# Set permissions
RUN chmod 600 /root/.ssh/id_rsa && \
    chmod 644 /root/.ssh/authorized_keys

# SSH login fix
RUN sed -i 's@session\\s*required\\s*pam_loginuid.so@session optional pam_loginuid.so@g' /etc/pam.d/sshd

EXPOSE 22

CMD ["/usr/sbin/sshd", "-D"]
```

Step-7:- Build Image

```
docker build -t ubuntu-server
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/ansible-lab$ docker build -t
 ubuntu-server .
[+] Building 36.7s (15/15) FINISHED                          docker:default
 => [internal] load build definition from Dockerfile                    0.1s
 => => transferring dockerfile: 947B                                    0.1s
 => [internal] load metadata for docker.io/library/ubuntu:latest        0.6s
 => [internal] load .dockerignore                                       0.0s
 => => transferring context: 2B                                         0.0s
 => [ 1/10] FROM docker.io/library/ubuntu:latest@sha256:186072bba1b2f   0.1s
 => => resolve docker.io/library/ubuntu:latest@sha256:186072bba1b2f43   0.1s
 => [internal] load build context                                       0.0s
 => => transferring context: 58B                                        0.0s
 => CACHED [ 2/10] RUN apt update -y                                    0.0s
 => CACHED [ 3/10] RUN apt install -y python3 python3-pip openssh-ser   0.0s
 => CACHED [ 4/10] RUN mkdir -p /var/run/sshd                           0.0s
 => CACHED [ 5/10] RUN mkdir -p /run/sshd &&     echo 'root:password'   0.0s
 => CACHED [ 6/10] RUN mkdir -p /root/.ssh &&     chmod 700 /root/.ss   0.0s
 => CACHED [ 7/10] COPY id_rsa /root/.ssh/id_rsa                        0.0s
 => CACHED [ 8/10] COPY id_rsa.pub /root/.ssh/authorized_keys           0.0s
 => CACHED [ 9/10] RUN chmod 600 /root/.ssh/id_rsa &&     chmod 644 /   0.0s
 => CACHED [10/10] RUN sed -i 's@session\\s*required\\s*pam_loginuid.   0.0s
 => exporting to image                                                 33.9s
 => => exporting layers                                                 0.0s
 => => exporting manifest sha256:840ae1782a62f55e094ffa4873746f023d01   0.0s
 => => exporting config sha256:ac809663b70999e5c3704934c5bb120e44ffda   0.0s
 => => exporting attestation manifest sha256:605cec12d055eaf93373c5b9   0.1s
 => => exporting manifest list sha256:63c3421f070f7291323fb9dafd10c6f   0.0s
 => => naming to docker.io/library/ubuntu-server:latest                 0.0s
 => => unpacking to docker.io/library/ubuntu-server:latest             33.6s
```

Step-8:- Run container from build Image

```
ocker run -d -p --rm 2222:22 -p 8221:8221 --name ssh-test-server ubuntu-server
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/ansible-lab$ docker run -d -
-rm -p 2222:22 --name ssh-test-server ubuntu-server
f9b11aaeaacecfa636be3b7c41b0e63c7f21217699e08471b3c5fab3b636c6d6
```

Step-9:- Find Container IP Address

```
docker inspect -f '' ssh-test-server
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/ansible-lab$ docker inspect
-f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' ssh-test-server
172.17.0.3
```

Step-10:- Test SSH connection via Conatiner IP

```
# with key
ssh -i ~/.ssh/id_rsa root@172.17.0.2
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/ansible-lab$ ssh -i ~/.ssh/i
d_rsa -p 2222 root@localhost
The authenticity of host '[localhost]:2222 ([127.0.0.1]:2222)' can't be esta
blished.
ED25519 key fingerprint is SHA256:QpwuqXurAQGdQi7hTNChbtgkebuz0VzTLe60Pq2zXi
k.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[localhost]:2222' (ED25519) to the list of known
 hosts.
Enter passphrase for key '/home/palakagarwal_16/.ssh/id_rsa':
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x8
6_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@69b9d4a2aea0:~# for i in {1..4}; do
  echo "Creating server${i}"
  docker run -d --rm -p 220${i}:22 --name server${i} ubuntu-server
  echo "IP of server${i} is $(docker inspect -f '{{range.NetworkSettings.Net
works}}{{.IPAddress}}{{end}}' server${i})"
done
Creating server1
-bash: docker: command not found
-bash: docker: command not found
IP of server1 is
```

Step-11:- Start Multiple Conatiners

```
for i in {1..4}; do
  echo -e "\n Creating server${i}\n"
  docker run -d --rm -p 220${i}:22 --name server${i} ubuntu-server
  echo -e "IP of server${i} is $(docker inspect -f '' server${i})"
done
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/ansible-lab$ for i in {1..4}
; do
  echo "Creating server${i}"
  docker run -d --rm -p 220${i}:22 --name server${i} ubuntu-server
  echo "IP of server${i} is $(docker inspect -f '{{range.NetworkSettings.Net
works}}{{.IPAddress}}{{end}}' server${i})"
done
Creating server1
e13b4852dc3ea7e7cb3aacae3778804dd44762c70e601880f1794d08a4f358bf
IP of server1 is 172.17.0.4
Creating server2
96f1627e29b73c3c73e5743811476dce1b085e128b91366dfa876cc8c4326f2d
IP of server2 is 172.17.0.5
Creating server3
1f720a4c0b912a8881fd49bb3e0f6c8895dec2691ab50d785bf4648e91ad7957
IP of server3 is 172.17.0.6
Creating server4
76ec9d4514811caeb2cd64cbc71e5548c25b1dcbbdb1ea407430d276b9e1edf6
IP of server4 is 172.17.0.7
```

Step-12:- Create `inventory.ini`

```
# Get container IPs
echo "[servers]" > inventory.ini
for i in {1..4}; do
  docker inspect -f '' server${i} >> inventory.ini
done

# Add inventory variables
cat << EOF >> inventory.ini

[servers:vars]
ansible_user=root
; ansible_ssh_private_key_file=./id_rsa
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
EOF
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/ansible-lab$ echo "[servers]
" > inventory.ini

for i in {1..4}; do
  port=220${i}
  echo "server${i} ansible_host=127.0.0.1 ansible_port=${port}" >> inventory
.ini
done

cat << EOF >> inventory.ini

[servers:vars]
ansible_user=root
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
EOF
```

Step-13:- Verify `inventory.ini`

```
cat inventory.ini
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/ansible-lab$ cat inventory.i
ni
[servers]
server1 ansible_host=127.0.0.1 ansible_port=2201
server2 ansible_host=127.0.0.1 ansible_port=2202
server3 ansible_host=127.0.0.1 ansible_port=2203
server4 ansible_host=127.0.0.1 ansible_port=2204

[servers:vars]
ansible_user=root
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
```

Step-14:- Test Connectivity

```
ssh -i ~/.ssh/id_rsa root@172.17.0.3
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/ansible-lab$ ssh -i ~/.ssh/i
d_rsa -p 2201 root@localhost
The authenticity of host '[localhost]:2201 ([127.0.0.1]:2201)' can't be esta
blished.
ED25519 key fingerprint is SHA256:QpwuqXurAQGdQi7hTNChbtgkebuz0VzTLe60Pq2zXi
k.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[localhost]:2201' (ED25519) to the list of known
 hosts.
Enter passphrase for key '/home/palakagarwal_16/.ssh/id_rsa':
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x8
6_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@e13b4852dc3e:~# exit
logout
Connection to localhost closed.
```

Step-15:- Ansible Ping Test

```
ansible all -i inventory.ini -m ping
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/ansible-lab$ ANSIBLE_HOST_KE
Y_CHECKING=False ansible all -i inventory.ini -m ping
[WARNING]: Ansible is being run in a world writable directory
(/mnt/c/Users/hp/ansible-lab), ignoring it as an ansible.cfg source. For mor
e
information see
https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-
in-
world-writable-dir
server2 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
server1 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
server4 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
server3 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/ansible-lab$
```

Step-16:- Create Playbook `playbook1.yml`

```yaml
--- # it should start with three dash only
- name: Update and configure servers
  hosts: all
  become: yes

  tasks:
    - name: Update apt packages
      apt:
        update_cache: yes
        upgrade: dist

    - name: Install required packages
      apt:
        name: ["vim", "htop", "wget"]
        state: present

    - name: Create test file
      copy:
        dest: /root/ansible_test.txt
        content: "Configured by Ansible on "
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/ansible-lab$ nano playbook1.
yml
```

Step-17:- Run Playbook

```
ansible-playbook -i inventory.ini playbook1.yml
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/ansible-lab$ ansible-playboo
k -i inventory.ini playbook1.yml
[WARNING]: Ansible is being run in a world writable directory
(/mnt/c/Users/hp/ansible-lab), ignoring it as an ansible.cfg source. For mor
e
information see
https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-
in-
world-writable-dir

PLAY [Update and configure servers] ***************************************
****

TASK [Gathering Facts] ****************************************************
****
ok: [server2]
ok: [server3]
ok: [server1]
ok: [server4]

TASK [Update apt packages] ************************************************
****
changed: [server3]
changed: [server1]
changed: [server4]
changed: [server2]

TASK [Install required packages] ******************************************
****
changed: [server1]
changed: [server3]
changed: [server4]
changed: [server2]

TASK [Create test file] ***************************************************
****
changed: [server4]
changed: [server3]
changed: [server1]
```

Step-18:- Verify Changes

```
ansible all -i inventory.ini -m command -a "cat /root/ansible_test.txt"
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/ansible-lab$ ansible all -i
inventory.ini -m command -a "cat /root/ansible_test.txt"
[WARNING]: Ansible is being run in a world writable directory
(/mnt/c/Users/hp/ansible-lab), ignoring it as an ansible.cfg source. For mor
e
information see
https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-
in-
world-writable-dir
server1 | CHANGED | rc=0 >>
Configured by Ansible on server1
server3 | CHANGED | rc=0 >>
Configured by Ansible on server3
server4 | CHANGED | rc=0 >>
Configured by Ansible on server4
server2 | CHANGED | rc=0 >>
Configured by Ansible on server2
```

Step-19:- CleanUp

```
for i in {1..4}; do docker rm -f server${i}; done
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/ansible-lab$ for i in {1..4}
; do docker rm -f server${i}; done
server1
server2
server3
server4
```